

PINGCHENG JIAN

PhD student, Electrical and Computer Engineering, Duke University

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EDUCATION

Duke University

Ph.D. in Electrical and Computer Engineering

Sep. 2021 - present

Advisors: [Michael Zavlanos](#), [Boyuan Chen](#)

Tsinghua University

B.E. in Automation

Transfer major from Mechanical Engineering (2016-2018)
to Automation (2018-2021)

Aug. 2016 - Jul. 2021

Advisors: [Huaping Liu](#), [Mingguo Zhao](#)

RESEARCH INTERESTS

My current research is about robot learning, and my previous publications mostly focus on robotic manipulation. My current research focuses on applying the foundation models to robotic tasks.

PUBLICATIONS

(* equal contributions, ** alphabetical order. Up-to-date list can be found in my Google Scholar page.)

Journal Papers

1. [TMLR'24] Pingcheng Jian, Easop Lee, Zachary I. Bell, Michael M. Zavlanos, Boyuan Chen, "Perception Stitching: Zero-Shot Perception Encoder Transfer for Visuomotor Robot Policies", *Transactions on Machine Learning Research (TMLR)*, 2024. [[openreview](#)] [[website](#)] [[video](#)] [[arxiv](#)]

Conference Papers

2. [CoRL'23] Pingcheng Jian, Easop Lee, Zachary Bell, Michael M. Zavlanos, Boyuan Chen, "Policy Stitching: Learning Transferable Robot Policies", *Conference on Robot Learning (CoRL)*, 2023. [[website](#)] [[CoRL talk](#)] [[video](#)] [[arXiv](#)]
3. [ICRA'21] Pingcheng Jian*, Chao Yang*, Di Guo, Huaping Liu, Fuchun Sun, "Adversarial Skill Learning for Robust Manipulation", *International Conference on Robotics and Automation (ICRA)*, 2021. [[pdf](#)] [[video](#)]
4. [L4DC'23] Reza Khodayi-mehr, Pingcheng Jian, Michael M. Zavlanos, "Physics-Guided Active Learning of Environmental Flow Fields", *Learning for Dynamics and Control (L4DC)*, 2023. [[pdf](#)]

Workshop Papers

5. [ICML-W'21] Minghao Zhang*, Pingcheng Jian*, Yi Wu, Huazhe Xu, Xiaolong Wang, "DAIR: Disentangled Attention Intrinsic Regularization for Safe and Efficient Bimanual Manipulation", *International Conference on Machine Learning (ICML) Workshop: Reinforcement Learning for Real Life*, 2021. [[website](#)]

PROJECTS

Quadruped Spider Robot, course project of CS590 - *Robot Studio* at Duke University

Project Video: [[video](#)]

I built all the hardware and software of this spider robot from scratch. I drew the parts of this robot

with SOLIDWORKS and then printed them with 3D printer. The controller of this robot is a Raspberry Pi, and the control algorithm is written in Python. Then I generated the .urdf files of this robot and built the simulator of this robot with PyBullet.

ACADEMIC SERVICE

Reviewer of Conference on Robot Learning (CoRL), 2024

TEACHING EXPERIENCES

Teaching Assistant at Duke University

[ECE 382L/ME 344L](#) - Control of Dynamic Systems [\[lecture\]](#) [\[recitation\]](#)

[ECE 689/COMPSCI 676](#) - Advanced Topics in Deep Learning

ADVISING AND MENTORSHIP

Easop Lee, Duke Undergraduate Student in ECE (2022). Next: Ph.D. at Duke ECE

Xiao Wei, Duke CS Master (2024 - present)

Yongjae Lee, Duke ME Master (2024 - present)

SKILLS

Programming Languages

Python, Matlab, C, C++, C#, HTML, Assembly Language

Machine Learning

Pytorch, Reinforcement Learning, Supervised Learning

Robotic Software

PyBullet, MuJoCo, Isaac Gym

Robotic Hardware

SOLIDWORKS, 3D printing